AKTIVITAS MANDIRI 3 — Membuat Layout Master & Halaman-Halaman

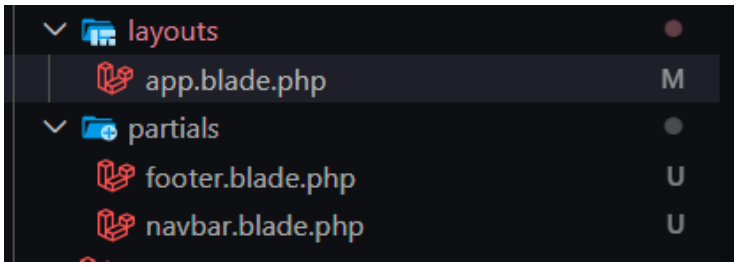
1. Buat layout master: resources/views/layouts/app.blade.php dengan navbar, main content area (@yield('content')), dan footer

```
CAKEYS > resources > views > layouts > app.blade.php > ...
1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta name="viewport" content="width=device-width, initial-scale=1.0">
6      <meta http-equiv="X-UA-Compatible" content="ie=edge">
7      <title>@yield('title', 'Cakeys - Dashboard Owner')</title>
8      <link rel="stylesheet" href="{{ asset('style.css') }}">
9      <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700;800&display=swap" rel="stylesheet">
10 </head>
11 <body>
12     <div class="menu-overlay" id="menuOverlay"></div>
13     @include('partials.navbar')
14
15     <main class="main-content {{--dashboard-wrapper--}}">
16         @yield('content')
17     </main>
18
19     @include('partials.footer')
20
21 </body>
22 </html>
23
```

Di sini saya membuat app.blade.php yang nantinya berfungsi sebagai kerangka utama (master layout) website yang memuat aset global, menyisipkan elemen tetap seperti navbar dan footer menggunakan perintah @include, serta menyediakan ruang dinamis melalui @yield('content') untuk diisi oleh konten spesifik dari halaman-halaman turunannya.

2. Buat partial: resources/views/partials/navbar.blade.php dan sisipkan dengan @include di layout

Di sini saya membuat file navbar.blade.php dan footer.blade.php di dalam direktori partials untuk memisahkan kode navigasi dan footer menjadi komponen modular, sehingga kode menjadi lebih rapi dan dapat dengan mudah disisipkan ke dalam master layout menggunakan perintah @include (seperti yang ada di screenshot kode nomor 1).



CAKEYS > resources > views > partials > navbar.blade.php > nav.navbar > div.nav-icon

```
<nav class="navbar">
    <div class="nav-kiri">
        <img src="{{ asset('images/opsi-logo-2.png') }}" alt="logo" class="logo">

        <div class="nav-teks">
            <h1>CAKEYS</h1>
            <p>Create Your Own Cake</p>
        </div>
    </div>

    <ul class="nav-menu">
        <li><a href="{{ route('dashboard') }}">Dashboard</a></li>
        <li><a href="{{ route('katalog') }}">Katalog</a></li>
        <li><a href="{{ route('pesanan') }}">Pesanan</a></li>
        <li><a href="{{ route('profile') }}">Profile</a></li>
    </ul>

    <div class="nav-icon">
        <div class="search-container" id="searchContainer">
            <span class="icon search-btn bitcoin-icons--search-outline" id="searchBtn"></span>
            <form class="search-box" id="searchBox" onsubmit="event.preventDefault();">
                <input type="text" id="searchInput" placeholder="Cari pesanan/produk...">
            </form>
        </div>

        <div class="profile-container" id="profileContainer">
            <span class="icon healthicons--ui-user-profile" id="profileBtn"></span>
            <div class="dropdown" id="profileDropdown">
                <p><strong>Halo, Owner!</strong></p>
                <hr style="margin: 5px 0; border: 0; border-top: 1px solid #ddd;">
                <p><a href="{{ route('profile') }}" style="color: inherit; text-decoration: none;">Profil Saya</a></p>
                <p>Pengaturan</p>
                <p class="logout-btn" {{--style="color: #E11D48;"--}}>Logout</p>
            </div>
        </div>
    </div>
</nav>
```

CAKEYS > resources > views > partials > footer.blade.php > ...

```
<footer class="footer">
    <div class="footer-container">

        <div class="footer-brand">
            <div class="footer-logo">
                <img src="{{ asset('images/opsi-logo-2.png') }}" alt="Logo Cakeys">
                <div>
                    <h2>CAKEYS</h2>
                    <p>Create Your Own Cake</p>
                </div>
            </div>
            <p>
                Toko kue custom untuk berbagai acara. Nikmati acara dengan kue spesial sesuai keinginanmu! 🎂
            </p>
        </div>

        <div class="footer-center">
            <h3>Menu</h3>
            <ul>
                <li><a href="{{ route('dashboard') }}">Dashboard</a></li>
                <li><a href="{{ route('katalog') }}">Katalog</a></li>
                <li><a href="{{ route('pesanan') }}">Pesanan</a></li>
                <li><a href="{{ route('profile') }}">Profile</a></li>
            </ul>
        </div>

        <div>
            <h3>Hubungi Kami</h3>
            <p>📍 Alamat: Bondowoso, Jawa Timur</p>
            <p>📞 Telp: 0812-3456-7890</p>
            <p>📷 Instagram: @cakeyscake</p>
            <p>✉️ Email: cakeys@gmail.com</p>
        </div>

    </div>

    <div>
        <p>© 2026 Cakeys | Naila's Project</p>
    </div>
</footer>
```

3. Buat halaman dashboard: resources/views/dashboard.blade.php yang @extends('layouts.app')

Di sini saya membuat halaman dashboard.blade.php yang menggunakan perintah @extends('layouts.app') untuk mewarisi struktur master layout, serta memanfaatkan @section('content') untuk menampilkan desain konten spesifik, seperti data overview produk dan ringkasan pesanan ke dalam ruang dinamis yang telah disediakan pada kerangka utama.

```
CAKEYS > resources > views > dashboard.blade.php > ...
1  @extends('layouts.app')
2
3  @section('title', 'Dashboard Owner - Cakeys')
4
5  @section('content')
6  <div class="dashboard-wrapper">
7      <div class="dashboard-welcome">
8          <h3>Cakeys's Dashboard</h3>
9          <h1>Halo, {{ $username }}! Selamat Datang...</h1>
10     </div>
11
12     <div class="dashboard-section mt-4">
13         <h2>Overview Produk</h2>
14         <div class="grid-container">
15             <div class="card">
16                 <h3>Total Produk</h3>
17                 <p class="angka">{{ $overview['total_produk'] }}</p>
18             </div>
19
20             <div class="card">
21                 <h3>Produk Terlaris</h3>
22                 <p class="teks">{{ $overview['produk_terlaris'] }}</p>
23             </div>
24
25             <div class="card">
26                 <h3>Stok Menipis</h3>
27                 <p class="angka">{{ $overview['stok_menipis'] }} Produk</p>
28             </div>
29         </div>
30     </div>
31
32     <div class="dashboard-section">
33         <h2>Ringkasan Pesanan</h2>
34         <div class="grid-container">
35             <div class="card">
36                 <h3>Total Order</h3>
37                 <p class="angka">{{ $ringkasan_pesanan['total_order'] }}</p>
38             </div>
39
40             <div class="card">
41                 <h3>Produk Diorder</h3>
42                 <p class="angka">{{ $ringkasan_pesanan['total_produkorder'] }}</p>
43             </div>
44
45             <div class="card-status">
```

127.0.0.1:8000/dashboard

CAKEYS
Create Your Own Cake

Dashboard Katalog Pesanan Profile

Cakeys's Dashboard

Halo, Owner! Selamat Datang...

Overview Produk

Total Produk	Produk Terlaris	Stok Menipis
13	Chocolate Strawberry Cake	1 Produk

Ringkasan Pesanan

Total Order	Produk Diorder	Diproses	Dikirim	Selesai	Total Pendapatan
5	9	1	2	1	Rp 4.750.000

Orderan Masuk Terbaru

Lihat Semua

CAKEYS
Create Your Own Cake

Dashboard Katalog Pesanan Profile

Cakeys's Dashboard

Halo, Owner! Selamat Datang...

Overview Produk

Total Produk	Produk Terlaris	Stok Menipis
13	Chocolate Strawberry Cake	1 Produk

Ringkasan Pesanan

Total Order	Produk Diorder	Diproses	Dikirim	Selesai	Total Pendapatan
5	9	1	2	1	Rp 4.750.000

Orderan Masuk Terbaru

Lihat Semua

ID Order	Tanggal	Customer	Nama Produk	Subtotal	Status Pesanan
P001	24-04-26	Esther 081234567890	Fudgy Chocolate Brownies (x2)	Rp 600.000	Diproses
P002	24-04-26	Ophelia 081325476980	Tiramisu Cake (x2)	Rp 1.000.000	Dikirim
P003	24-04-26	Amelia 08237465819	Chocolate Cream Pie (x1)	Rp 500.000	Dikirim
P004	24-04-26	Sabil 08134675281	Apple Pie (x1)	Rp 520.000	Diproses
P005	24-04-26	Michell 085326756456	Chocolate Strawberry Cake (x3)	Rp 1.950.000	Selesai

CAKEYS
Create Your Own Cake

Toko kue custom untuk berbagai acara. Nikmati acara dengan kue spesial sesuai keinginanmu! 🎂

Menu

- Dashboard
- Katalog
- Pesanan
- Profile

Hubungi Kami

📍 Alamat: Bondowoso, Jawa Timur
📞 Telp: 0812-3456-7890
📷 Instagram: @cakeyscake
✉ Email: cakeys@gmail.com

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4. Tampilkan flash session message di layout: @if(session('success')) ... @endif

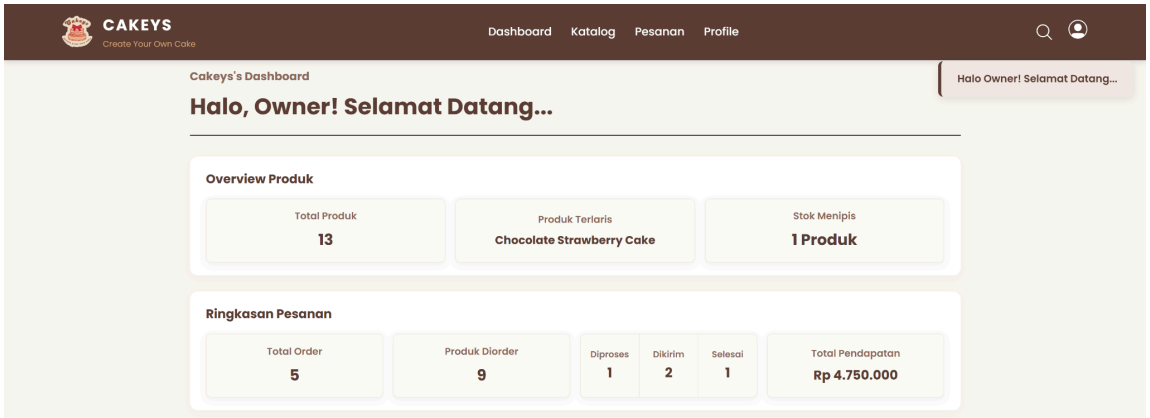
Di sini saya menambahkan blok logika @if(session('success')) pada master layout (app.blade.php) untuk menangkap data flash session yang dikirimkan dari controller, sehingga website dapat menampilkan notifikasi pop-up interaktif (seperti ucapan selamat datang) yang akan menghilang secara otomatis kepada pengguna.

CAKEYS > resources > views > layouts > app.blade.php > ...

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <meta http-equiv="X-UA-Compatible" content="ie=edge">
    <title>@yield('title', 'Cakeys - Dashboard Owner')</title>
    <link rel="stylesheet" href="{{ asset('style.css') }}">
    <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@400;500;600;700;800&display=swap" rel="stylesheet">
</head>
<body>
    <div class="menu-overlay" id="menuOverlay"></div>
    @include('partials.navbar')

    @if(session('success'))
        <div id="flash-message" style="position: fixed; top: 90px; right: 20px; background-color: #f4e8e1; color: #5A3E36;
            {{ session('success') }}
        </div>

        <script>
            setTimeout(function() {
                let flashMessage = document.getElementById('flash-message');
                if(flashMessage) {
                    flashMessage.style.opacity = '0';
                    setTimeout(() => flashMessage.remove(), 500);
                }
            }, 3000);
        </script>
    @endif

    <main class="main-content {{--dashboard-wrapper--}}">
        @yield('content')
    </main>

    @include('partials.footer')

</body>
</html>
```

CAKEYS > app > Http > Controllers > DashboardController.php > DashboardController > index

```
class DashboardController extends Controller
    public function index()
        session()->flash('success', 'Halo Owner! Selamat Datang...');

        return view('dashboard', compact('username', 'ringkasan_pesanan', 'overview', 'order'));
    }
}
```



5. Buat komponen: resources/views/components/stat-card.blade.php (kartu statistik dengan props: judul, nilai, ikon, warna)

   Di sini saya membuat komponen reusable pada file stat-card.blade.php yang memanfaatkan perintah @props untuk menerima data dinamis seperti judul, nilai, ikon, dan warna, sehingga elemen kartu statistik dapat dipanggil berkali-kali secara efisien tanpa harus menduplikasi kode HTML-nya.

```
CAKEYS > resources > views > components > stat-card.blade.php > ...
1  @props([
2      'judul',
3      'nilai',
4      'ikon' => '',
5      'warna' => 'var(--primary-color)'
6  ])
7
8  <div class="card">
9      <h3>
10         @if($ikon)
11             <span style="margin-right: 5px;">{!! $ikon !!}</span>
12         @endif
13         {{ $judul }}
14     </h3>
15     <p class="angka" style="color: {{ $warna }};">{{ $nilai }}</p>
16 </div>
17
```

6. Gunakan @forelse di dashboard untuk loop array data dummy (jumlah mahasiswa, mata kuliah, dll)

   Di sini saya menggunakan perintah @forelse pada halaman dashboard untuk melakukan perulangan data pada array overview, ringkasan pesanan, dan daftar orderan terbaru. Setiap datanya ditampilkan menggunakan komponen stat-card.blade.php yang sudah dibuat, serta menyediakan tampilan cadangan melalui blok @empty jika data tersebut kosong.

CAKEYS > resources > views > dashboard.blade.php > div.dashboard-wrapper > div.dashboard-section > div.table > table.table-order > tbody > tr

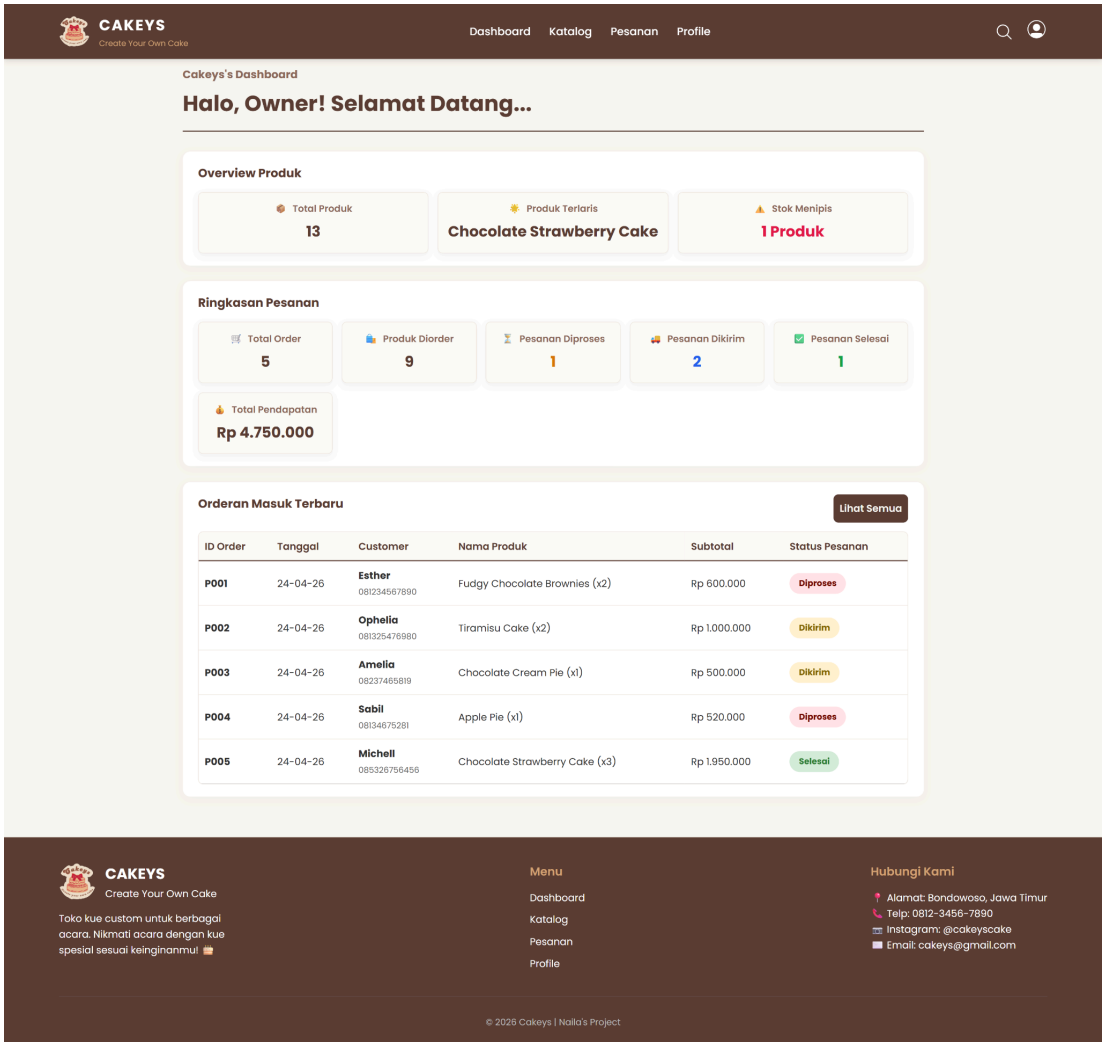
```
6   <div class="dashboard-wrapper">
32      <div class="dashboard-section mt-4">
33          <h2>Overview Produk</h2>
34          <div class="grid-container">
35              @forelse ($overview as $data)
36                  <x-stat-card
37                      :judul="$data['judul']"
38                      :nilai="$data['nilai']"
39                      :ikon="$data['ikon']"
40                      :warna="$data['warna']"
41                  />
42              @empty
43                  <p style="text-align: center; grid-column: span 3; color: #888;">Data statistik belum tersedia saat ini.</p>
44              @endforelse
45          </div>
46      </div>
```

CAKEYS > resources > views > dashboard.blade.php > div.dashboard-wrapper > div.dashboard-section > div.table > table.table-order > tbo

```
6   <div class="dashboard-wrapper">
85      <div class="dashboard-section">
86          <h2>Ringkasan Pesanan</h2>
87          <div class="grid-container">
88              @forelse ($ringkasan_pesanan as $data)
89                  <x-stat-card
90                      :judul="$data['judul']"
91                      :nilai="$data['nilai']"
92                      :ikon="$data['ikon']"
93                      :warna="$data['warna']"
94                  />
95              @empty
96                  <p style="text-align: center; grid-column: span 4;">Data ringkasan pesanan belum tersedia.</p>
97              @endforelse
98          </div>
99      </div>
```

CAKEYS > resources > views > dashboard.blade.php > div.dashboard-wrapper > div.dashboard-section > div.table > table.table-order > tbody > tr

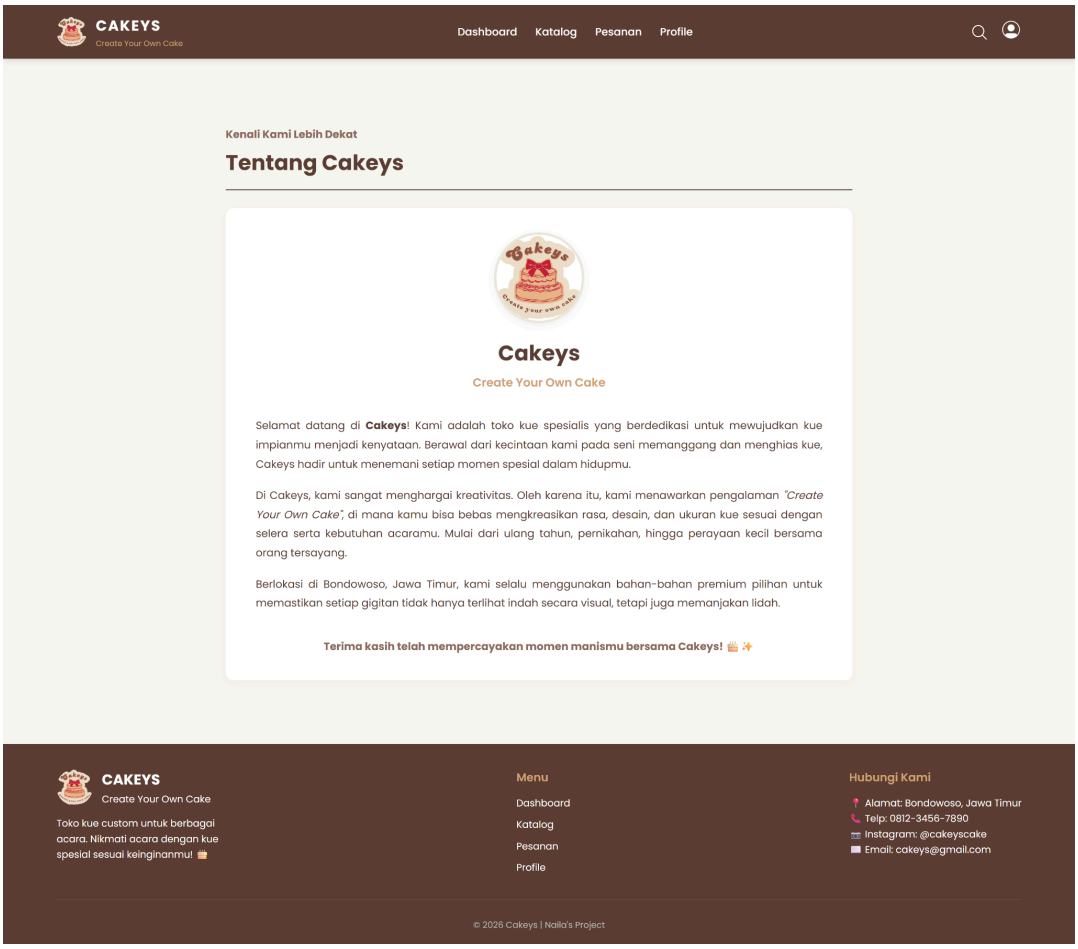
```
6   <div class="dashboard-wrapper">
101     <div class="dashboard-section">
108         <div class="table">
109             <table class="table-order">
147                 <tbody>
148                     @forelse ($order as $o)
149                         <tr style="border-bottom: 1px solid #ddd;">
150                             <td style="padding: 10px;"><strong>{{ $o['id_order'] }}</strong></td>
151                             <td style="padding: 10px;">{{ $o['tanggal_order'] }}</td>
152                             <td style="padding: 10px;">
153                                 <strong>{{ $o['nama_cust'] }}</strong><br>
154                                 <small style="color: #666;">{{ $o['no_telp'] }}</small>
155                             </td>
156                             <td style="padding: 10px;">{{ $o['nama_produk'] }} (x{{ $o['jumlah'] }})</td>
157                             <td style="padding: 10px;">Rp {{ number_format($o['subtotal'], 0, ',', '.') }}</td>
158                             <td style="padding: 10px;">
159                                 <span class="status-badge badge-{{ strtolower($o['status']) }}">
160                                     {{ $o['status'] }}
161                                 </span>
162                             </td>
163                         </tr>
164                     @empty
165                         <tr>
166                             <td colspan="6" style="padding: 20px; text-align: center; color: #888; font-style: italic;">
167                                 Belum ada orderan masuk saat ini.
168                             </td>
169                         </tr>
170                     @endforelse
171                 </tbody>
```



7. Buat halaman 'Tentang' dan 'Kontak' yang @extends layout yang sama
   a. Halaman Tentang

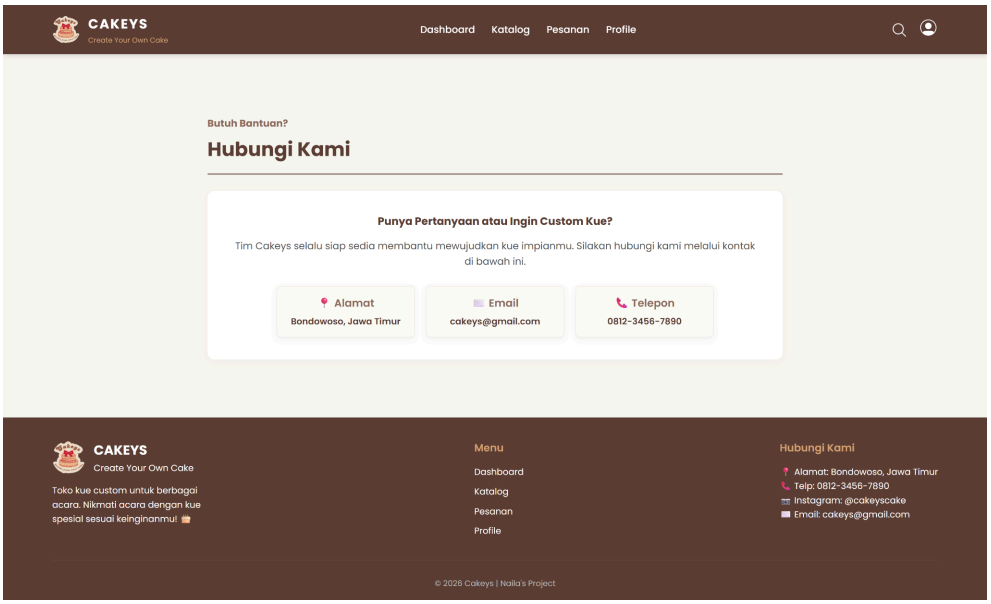
      Di sini saya membuat halaman tentang.blade.php dan kontak.blade.php yang menggunakan @extends('layouts.app') untuk mewarisi layout utama yang sama, sehingga kedua halaman tersebut memiliki tampilan navigasi dan footer yang konsisten dengan halaman lainnya pada website CAKEYS.

```
CAKEYS > resources > views > tentang.blade.php > ...
@extends('layouts.app')

@section('title', 'Tentang Kami - Cakeys')

@section('content')
<div class="main-content dashboard-wrapper">

    <div class="dashboard-welcome">
        <h3>Kenali Kami Lebih Dekat</h3>
        <h1>Tentang Cakeys</h1>
    </div>

    <div class="dashboard-section" style="padding: 40px 50px;">
        <div style="text-align: center; margin-bottom: 40px;">
            <img src="{{ asset('images/opsi-logo-2.png') }}" alt="Logo Cakeys" style="width: 150px; height: 150px; object-fit: cover; border-radius: 50%; margin-bo
            <h2 style="font-size: 2.2rem; color: var(--primary-color); margin-bottom: 5px;">Cakeys</h2>
            <p style="color: var(--accent-color); font-weight: 600; font-size: 1.2rem;">Create Your Own Cake</p>
        </div>

        <div style="line-height: 1.8; color: var(--primary-color); font-size: 1.1rem; text-align: justify;">
            <p style="margin-bottom: 20px;">
                Selamat datang di <strong>Cakeys</strong>! Kami adalah toko kue spesialis yang berdedikasi untuk mewujudkan kue impianmu menjadi kenyataan. Berawal
            </p>
            <p style="margin-bottom: 20px;">
                Di Cakeys, kami sangat menghargai kreativitas. Oleh karena itu, kami menawarkan pengalaman <em>"Create Your Own Cake"</em>, di mana kamu bisa bebas
            </p>
            <p style="margin-bottom: 20px;">
                Berlokasi di Bondowoso, Jawa Timur, kami selalu menggunakan bahan-bahan premium pilihan untuk memastikan setiap gigitan tidak hanya terlihat indah
            </p>
            <p style="text-align: center; font-weight: bold; margin-top: 40px; color: var(--secondary-color);">
                Terima kasih telah mempercayakan momen manismu bersama Cakeys! 🎂✨
            </p>
        </div>
    </div>

</div>
@endsection
```



b. Halaman Kontak

```
CAKEYS > resources > views > kontak.blade.php > ...
1  @extends('layouts.app')
2
3  @section('title', 'Kontak Kami - Cakeys')
4
5  @section('content')
6  <div class="main-content dashboard-wrapper">
7      <div class="dashboard-welcome">
8          <h3>Butuh Bantuan?</h3>
9          <h1>Hubungi Kami</h1>
10     </div>
11
12     <div class="dashboard-section" style="padding: 40px; text-align: center;">
13         <h2 style="color: var(--primary-color);">Punya Pertanyaan atau Ingin Custom Kue?</h2>
14         <p style="margin-bottom: 30px; font-size: 1.1rem; color: #555;">Tim Cakeys selalu siap sedia membantu mewujudkan kue impianmu.
15
16         <div style="display: flex; justify-content: center; gap: 20px; flex-wrap: wrap;">
17             <div class="card" style="width: 250px;">
18                 <h3 style="font-size: 1.2rem;">📍 Alamat</h3>
19                 <p style="font-weight: 600; color: var(--primary-color);">Bondowoso, Jawa Timur</p>
20             </div>
21             <div class="card" style="width: 250px;">
22                 <h3 style="font-size: 1.2rem;">✉️ Email</h3>
23                 <p style="font-weight: 600; color: var(--primary-color);">cakeys@gmail.com</p>
24             </div>
25             <div class="card" style="width: 250px;">
26                 <h3 style="font-size: 1.2rem;">📞 Telepon</h3>
27                 <p style="font-weight: 600; color: var(--primary-color);">0812-3456-7890</p>
28             </div>
29         </div>
30     </div>
31 </div>
32 @endsection
```

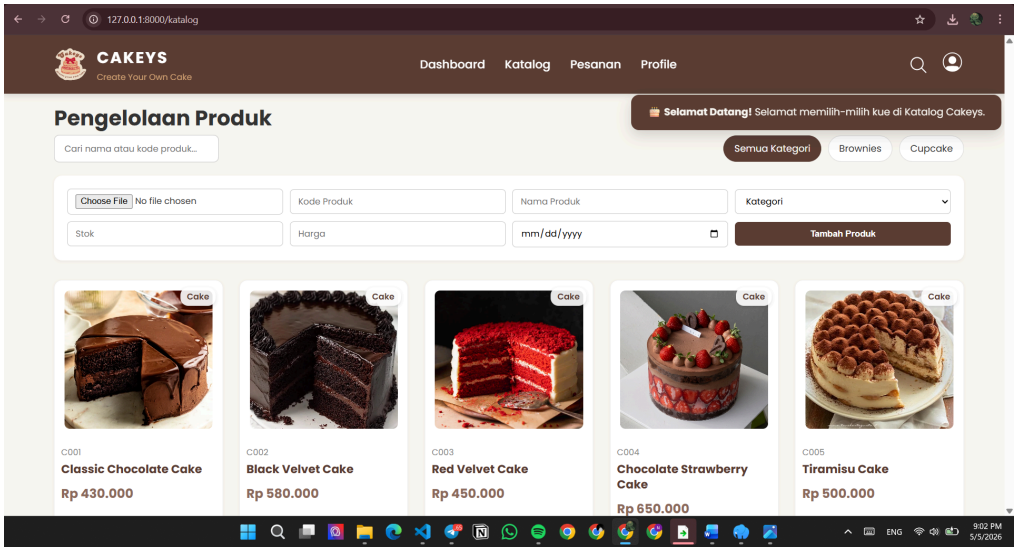


8. Gunakan @push('scripts') dan @stack('scripts') untuk menambahkan JavaScript spesifik per halaman

Di sini saya menggunakan perintah @stack('scripts') pada master layout sebagai tempat penampung, lalu menambahkan kode JavaScript khusus untuk notifikasi di halaman katalog menggunakan @push('scripts'), sehingga skrip tersebut hanya dimuat pada halaman tertentu tanpa mengganggu performa halaman lainnya.

```
CAKEYS > resources > views > layouts > app.blade.php > ...
2   <html lang="en">
11  <body>
37      @stack('scripts')
38  </body>
39  </html>
40
```

```
CAKEYS > resources > views > katalog.blade.php > div.katalog-container > div.top-bar > script > <function> > <function>
4   <div class="katalog-container">
6       <div class="top-bar">
70  @push('scripts')
71  <script>
72      document.addEventListener("DOMContentLoaded", function() {
73          let customAlert = document.createElement('div');
74          customAlert.innerHTML = "🎂 <strong>Selamat Datang!</strong> Selamat memilih-milih kue di Katalog Cakeys.";
75
76          Object.assign(customAlert.style, {
77              position: 'fixed',
78              top: '90px',
79              right: '20px',
80              backgroundColor: '#5A3E36',
81              color: '#F9F6F0',
82              padding: '15px 25px',
83              borderRadius: '12px',
84              boxShadow: '0 8px 20px rgba(0,0,0,0.15)',
85              fontFamily: "'Poppins', sans-serif",
86              fontSize: '0.95rem',
87              zIndex: '10000',
88              opacity: '0',
89              transform: 'translateY(-20px)',
90              transition: 'all 0.5s cubic-bezier(0.68, -0.55, 0.265, 1.55)'
91          });
92
93          document.body.appendChild(customAlert);
94
95          setTimeout(() => {
96              customAlert.style.opacity = '1';
97              customAlert.style.transform = 'translateY(0)';
98          }, 100);
99
100         setTimeout(() => {
101             customAlert.style.opacity = '0';
102             customAlert.style.transform = 'translateY(-20px)';
103             setTimeout(() => customAlert.remove(), 500);
104         }, 3500);
105     });
106 </script>
107 @endpush
```



💡 *Tips: Gunakan @include('partials.nama') untuk memecah tampilan menjadi komponen kecil yang reusable. Ini membuat kode View lebih mudah dikelola.*

AKTIVITAS MANDIRI 4 — Migration, Model & Seeder

1. Buat Model dan Migration sekaligus: php artisan make:model Mahasiswa -m

   Di sini saya menjalankan perintah php artisan make:model Produk -m untuk membuat file Model Produk.php sekaligus file migration secara bersamaan, yang mana Model akan berfungsi untuk mengelola logika data produk sedangkan migration berfungsi untuk merancang struktur tabelnya di database.

```
PS D:\NAII\PR PWEB\TUGAS TM> cd CAKEYS
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan make:model Produk -m

  INFO  Model [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\app\Models\Produk.php] created successfully.

  INFO  Migration [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\database\migrations\2026_05_05_142601_create_produks_table.php] created successfully.

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

2. Edit file migration: tambahkan kolom nim, nama, email, jurusan (enum), ipk (decimal), semester, aktif (boolean), foto (nullable)

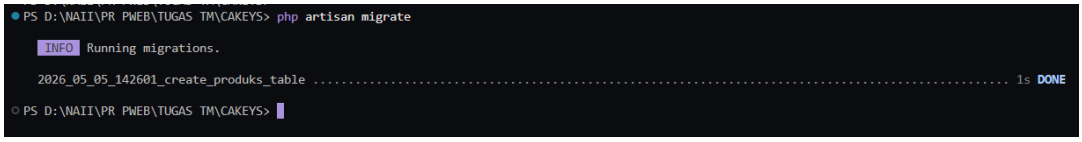
   Di sini saya mengubah file migration 2026_05_05_142601_create_produks_table.php dengan menambahkan kolom-kolom spesifik untuk tabel 'produks', seperti 'kode_produk' (string unique), 'nama_produk' (string), 'kategori' (enum untuk 'cake', 'brownies', 'cupcake'), 'harga' (decimal), 'stok' (integer), 'is_available' (boolean), serta 'gambar' yang bersifat nullable, guna mendefinisikan struktur data produk secara lengkap.

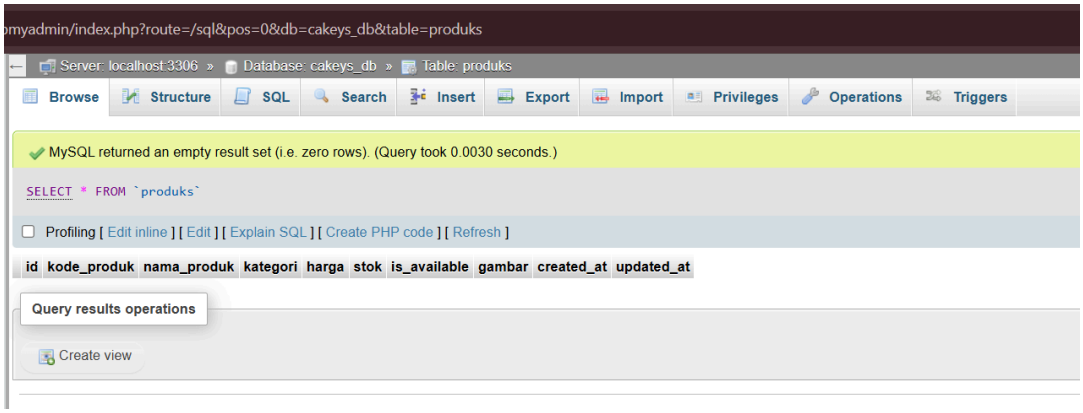
CAKEYS > database > migrations > 2026_05_05_142601_create_produks_table.php

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('produks', function (Blueprint $table) {
            $table->id();

            $table->string('kode_produk')->unique();
            $table->string('nama_produk');
            $table->enum('kategori', ['cake', 'brownies', 'cupcake']);
            $table->decimal('harga', 10, 2);
            $table->integer('stok');
            $table->boolean('is_available')->default(true);
            $table->string('gambar')->nullable();

            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('produks');
    }
};
```

3. Jalankan migration: php artisan migrate — cek di phpMyAdmin bahwa tabel sudah terbuat

Di sini saya menjalankan perintah php artisan migrate untuk mengeksekusi file migration yang telah dibuat sebelumnya, sehingga tabel produks beserta seluruh kolom yang telah didefinisikan otomatis terbuat di dalam database cakeys_db, sebagaimana terlihat pada tampilan phpMyAdmin.





4. Edit Model Mahasiswa: isi $fillable, tambahkan $casts, dan buat local scope scopeAktif()

Di sini saya melakukan konfigurasi pada Model Produk.php dengan mengisi properti $fillable untuk menentukan kolom mana saja yang boleh diisi secara massal, menambahkan $casts untuk memastikan tipe data is_available dan harga dikonversi secara otomatis, serta membuat local scope scopeTersedia() untuk memudahkan pemanggilan data produk yang statusnya tersedia dan stoknya masih ada.

```
CAKEYS > app > Models > Produk.php > ...
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Produk extends Model
{
    use HasFactory;

    protected $fillable = [
        'kode_produk',
        'nama_produk',
        'kategori',
        'harga',
        'stok',
        'is_available',
        'gambar'
    ];

    protected $casts = [
        'is_available' => 'boolean',
        'harga' => 'decimal:2',
    ];

    public function scopeTersedia($query)
    {
        return $query->where('is_available', true)->where('stok', '>', 0);
    }
}
```

5. Buat seeder: php artisan make:seeder MahasiswaSeeder — isi dengan 5-10 data dummy

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan make:seeder ProdukSeeder

  INFO  Seeder [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\database\seeders\ProdukSeeder.php] created successfully.

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

```
CAKEYS > database > seeders > ProdukSeeder.php > ProdukSeeder > run
namespace Database\Seeders;

use Illuminate\Database\Seeder;
use App\Models\Produk;

class ProdukSeeder extends Seeder
{
    public function run(): void
    {
        $dataProduk = [
            ['kode_produk' => 'C001', 'nama_produk' => 'Classic Chocolate Cake', 'kategori' => 'cake', 'harga' => 430000, 'stok' => 10, 'is_available' => true, 'ga
            ['kode_produk' => 'C002', 'nama_produk' => 'Black Velvet Cake', 'kategori' => 'cake', 'harga' => 580000, 'stok' => 10, 'is_available' => true, 'gambar'
            ['kode_produk' => 'C003', 'nama_produk' => 'Red Velvet Cake', 'kategori' => 'cake', 'harga' => 450000, 'stok' => 10, 'is_available' => true, 'gambar' =
            ['kode_produk' => 'C004', 'nama_produk' => 'Chocolate Strawberry Cake', 'kategori' => 'cake', 'harga' => 650000, 'stok' => 2, 'is_available' => true, '
            ['kode_produk' => 'C005', 'nama_produk' => 'Tiramisu Cake', 'kategori' => 'cake', 'harga' => 500000, 'stok' => 5, 'is_available' => true, 'gambar' => '
            ['kode_produk' => 'C006', 'nama_produk' => 'Cream Cheese Brownies', 'kategori' => 'brownies', 'harga' => 275000, 'stok' => 10, 'is_available' => true,
            ['kode_produk' => 'C007', 'nama_produk' => 'Fudgy Chocolate Brownies', 'kategori' => 'brownies', 'harga' => 300000, 'stok' => 8, 'is_available' => true
            ['kode_produk' => 'C008', 'nama_produk' => 'Strawberry Brownies', 'kategori' => 'brownies', 'harga' => 300000, 'stok' => 10, 'is_available' => true, 'g
            ['kode_produk' => 'C009', 'nama_produk' => 'Tiramisu Brownies', 'kategori' => 'brownies', 'harga' => 250000, 'stok' => 0, 'is_available' => false, 'gam
        ];

        foreach ($dataProduk as $produk) {
            Produk::create($produk);
        }
    }
}
```

6. Daftarkan seeder di DatabaseSeeder.php dan jalankan: php artisan db:seed

CAKEYS > database > seeders > DatabaseSeeder.php > ...

```php
namespace Database\Seeders;

use App\Models\User;
use Illuminate\Database\Console\Seeds\WithoutModelEvents;
use Illuminate\Database\Seeder;

class DatabaseSeeder extends Seeder
{
    /**
     * Seed the application's database.
     */
    public function run(): void
    {
        $this->call([
            ProdukSeeder::class,
        ]);
    }
}
```

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan db:seed

  INFO  Seeding database.

  Database\Seeders\ProdukSeeder ..................................................... RUNNING
  Database\Seeders\ProdukSeeder ..................................................... 729 ms DONE

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

Server: localhost:3306 » Database: cakeys_db » Table: produks

Browse | Structure | SQL | Search | Insert | Export | Import | Privileges | Operations | Triggers

Showing rows 0 - 8 (9 total, Query took 0.0045 seconds.)

```
SELECT * FROM `produks`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

	id	kode_produk	nama_produk	kategori	harga	stok	is_available	gambar	created_at	updated_at
Edit Copy Delete	1	C001	Classic Chocolate Cake	cake	430000.00	10	1	images/chocolate-cake-square.jpg	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	2	C002	Black Velvet Cake	cake	580000.00	10	1	images/black-velvet-cake-square.jpg	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	3	C003	Red Velvet Cake	cake	450000.00	10	1	images/red-velvet-cake-square.jpg	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	4	C004	Chocolate Strawberry Cake	cake	650000.00	2	1	images/choco-strawberry-square.png	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	5	C005	Tiramisu Cake	cake	500000.00	5	1	images/tiramisu-cake-square.jpg	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	6	C006	Cream Cheese Brownies	brownies	275000.00	10	1	images/cream-cheese-brownies.png	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	7	C007	Fudgy Chocolate Brownies	brownies	300000.00	8	1	images/fudgy-chocolate-brownies.jpg	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	8	C008	Strawberry Brownies	brownies	300000.00	10	1	images/strawberry-brownies.webp	2026-05-05 14:58:30	2026-05-05 14:58:30
Edit Copy Delete	9	C009	Tiramisu Brownies	brownies	250000.00	0	0	images/tiramisu-brownie.webp	2026-05-05 14:58:30	2026-05-05 14:58:30

7. Buka php artisan tinker dan coba query: Mahasiswa::all(), Mahasiswa::where('jurusan','SI')->get(), Mahasiswa::avg('ipk')

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan db:seed
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan tinker
Psy Shell v0.12.22 (PHP 8.3.30 — cli) by Justin Hileman
New PHP manual is available (latest: 3.0.5). Update with `doc --update-manual`

> App\Models\Produk::all()

= Illuminate\Database\Eloquent\Collection {#8080
    all: [
      App\Models\Produk {#8071
        id: 1,
        kode_produk: "C001",
        nama_produk: "Classic Chocolate Cake",
        kategori: "cake",
        harga: "430000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/chocolate-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8070
        id: 2,
        kode_produk: "C002",
        nama_produk: "Black Velvet Cake",
        kategori: "cake",
        harga: "580000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/black-velvet-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8073
        id: 3,
        kode_produk: "C003",
        nama_produk: "Red Velvet Cake",
        kategori: "cake",
        harga: "450000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/red-velvet-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8074
        id: 4,
        kode_produk: "C004",
        nama_produk: "Chocolate Strawberry Cake",
        kategori: "cake",
```

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan tinker
> App\Models\Produk::where('kategori', 'brownies')->get()

= Illuminate\Database\Eloquent\Collection {#8068
    all: [
      App\Models\Produk {#8088
        id: 6,
        kode_produk: "C006",
        nama_produk: "Cream Cheese Brownies",
        kategori: "brownies",
        harga: "275000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/cream-cheese-brownies.png",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8067
        id: 7,
        kode_produk: "C007",
        nama_produk: "Fudgy Chocolate Brownies",
        kategori: "brownies",
        harga: "300000.00",
        stok: 8,
        is_available: 1,
        gambar: "images/fudgy-chocolate-brownies.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#7816
        id: 8,
        kode_produk: "C008",
        nama_produk: "Strawberry Brownies",
        kategori: "brownies",
        harga: "300000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/strawberry-brownies.webp",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#7361
        id: 9,
        kode_produk: "C009",
        nama_produk: "Tiramisu Brownies",
        kategori: "brownies",
        harga: "250000.00",
        stok: 0,
        is_available: 0,
        gambar: "images/tiramisu-brownie.webp",
```

```
> App\Models\Produk::avg('harga')

= "415000.000000"
```

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan tinker
> App\Models\Produk::tersedia()->get()

= Illuminate\Database\Eloquent\Collection {#8089
    all: [
      App\Models\Produk {#8091
        id: 1,
        kode_produk: "C001",
        nama_produk: "Classic Chocolate Cake",
        kategori: "cake",
        harga: "430000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/chocolate-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8090
        id: 2,
        kode_produk: "C002",
        nama_produk: "Black Velvet Cake",
        kategori: "cake",
        harga: "580000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/black-velvet-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8093
        id: 3,
        kode_produk: "C003",
        nama_produk: "Red Velvet Cake",
        kategori: "cake",
        harga: "450000.00",
        stok: 10,
        is_available: 1,
        gambar: "images/red-velvet-cake-square.jpg",
        created_at: "2026-05-05 14:58:30",
        updated_at: "2026-05-05 14:58:30",
      },
      App\Models\Produk {#8069
        id: 4,
        kode_produk: "C004",
        nama_produk: "Chocolate Strawberry Cake",
        kategori: "cake",
        harga: "650000.00",
        stok: 2,
        is_available: 1,
        gambar: "images/choco-strawberry-square.png",
```

```
> exit

  INFO  Goodbye.

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

8. Coba buat relasi: buat tabel mata_kuliahs dan tabel pivot krs, lalu definisikan relasi belongsToMany di Model

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan make:model Topping -m

  INFO  Model [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\app\Models\Topping.php] created successfully.

  INFO  Migration [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\database\migrations\2026_05_05_152526_create_toppings_table.php] created successfully.

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

CAKEYS > database > migrations > 2026_05_05_152526_create_toppings_table.php

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('toppings', function (Blueprint $table) {
            $table->id();
            $table->string('nama_topping');
            $table->decimal('tambahan_harga', 10, 2);
            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('toppings');
    }
};
```

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan make:migration create_produk_topping_table

  INFO  Migration [D:\NAII\PR PWEB\TUGAS TM\CAKEYS\database\migrations\2026_05_05_153537_create_produk_topping_table.php] created successfully.

PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS>
```

CAKEYS > database > migrations > 2026_05_05_153537_create_produk_topping_table.php

```php
<?php

use Illuminate\Database\Migrations\Migration;
use Illuminate\Database\Schema\Blueprint;
use Illuminate\Support\Facades\Schema;

return new class extends Migration
{
    /**
     * Run the migrations.
     */
    public function up(): void
    {
        Schema::create('produk_topping', function (Blueprint $table) {
            $table->id();

            $table->foreignId('produk_id')->constrained('produks')->onDelete('cascade');

            $table->foreignId('topping_id')->constrained('toppings')->onDelete('cascade');

            $table->timestamps();
        });
    }

    /**
     * Reverse the migrations.
     */
    public function down(): void
    {
        Schema::dropIfExists('produk_topping');
    }
};
```

```
PS D:\NAII\PR PWEB\TUGAS TM\CAKEYS> php artisan migrate

  INFO  Running migrations.

  2026_05_05_152526_create_toppings_table ..... 977.67ms DONE
  2026_05_05_153537_create_produk_topping_table ..... 926.62ms DONE
```

Server: localhost:3306 » Database: cakeys_db » Table: toppings

Browse | Structure | SQL | Search | Insert | Exp

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0178 seconds

```
SELECT * FROM `toppings`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

id	nama_topping	tambahan_harga	created_at	updated_at

Query results operations

Create view

Server: localhost:3306 » Database: cakeys_db » Table: produk_topping

Browse | Structure | SQL | Search | Insert | Export

MySQL returned an empty result set (i.e. zero rows). (Query took 0.0066 seconds.)

```
SELECT * FROM `produk_topping`
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

id	produk_id	topping_id	created_at	updated_at

Query results operations

Create view

CAKEYS > app > Models > Produk.php > Produk > toppings

```
8   class Produk extends Model
32      public function toppings()
33      {
34          return $this->belongsToMany(Topping::class, 'produk_topping');
35      }
36  }
37
```

CAKEYS > app > Models > Topping.php > ...

```php
<?php

namespace App\Models;

use Illuminate\Database\Eloquent\Factories\HasFactory;
use Illuminate\Database\Eloquent\Model;

class Topping extends Model
{
    use HasFactory;

    protected $fillable = [
        'nama_topping',
        'tambahan_harga'
    ];

    public function produks()
    {
        return $this->belongsToMany(Produk::class, 'produk_topping');
    }
}
```

💡 *Tips: Di Tinker, bisa langsung uji query Eloquent tanpa membuat halaman. Sangat berguna untuk eksplorasi dan debugging.*